THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · EASY LEVEL CHAPTER TEST SET 3

Chapter 9: **Symmetry** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of these has no line of symmetry? 1

(a) O **(b)** A **(c)** H **(d)** L

2. The angle of rotation of a figure with rotational symmetry of order 6 is (degrees): 1

(a) 50 **(b)** 70 **(c)** 90 **(d)** 60

3. An isosceles triangle has how many lines of symmetry? 1

(a) 1 **(b)** 2 **(c)** 0 **(d)** 3

4. A figure is rotated through a full turn. It coincides with itself how many times in one full turn if its order of rotational symmetry is 6? 1

(a) 8 **(b)** 5 **(c)** 6 **(d)** 7

5. The number of lines of symmetry of a regular octagon is: 1

(a) 10 **(b)** 8 **(c)** 7 **(d)** 9

6. The number of lines of symmetry of the letter X (in capital) is: 1

(a) 3 **(b)** 2 **(c)** 1 **(d)** 4

7. Which of these has no line of symmetry? 1

(a) P **(b)** A **(c)** O **(d)** H

8. The angle of rotation of a figure with rotational symmetry of order 8 is (degrees): 1

(a) 35 **(b)** 45 **(c)** 75 **(d)** 55

9. The mirror image of a figure is also called its: 1

(a) rotation **(b)** perimeter **(c)** reflection **(d)** area

10. A figure is rotated through a full turn. It coincides with itself how many times in one full turn if its order of rotational symmetry is 12? 1

(a) 14 **(b)** 11 **(c)** 13 **(d)** 12

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Which of the letters B, F, T, X, M have a vertical line of symmetry? 2

12. How many lines of symmetry does a regular pentagon have? 2

13. Write the angle of rotational symmetry of a regular octagon. 2

14. Which of the letters F, T, X, M, G have a vertical line of symmetry? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. State the number of lines of symmetry and order of rotational symmetry of a regular octagon. 3

16. Classify each letter in O, S, X, Z as having a line of symmetry (yes/no). 3

17. Draw (describe) the mirror image of the point at 6 units to the left of a vertical mirror line. How far is its image from the mirror? 3

OR

State the number of lines of symmetry and order of rotational symmetry of a regular decagon. 3

18. Classify each letter in S, X, Z, A as having a line of symmetry (yes/no). 3

OR

Draw (describe) the mirror image of the point at 7 units to the left of a vertical mirror line. How far is its image from the mirror? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. A regular polygon has 8 sides. Write its number of lines of symmetry, its order of rotational symmetry and its smallest angle of rotation. Name the polygon. 5

OR

Which of the letters N, S, I, A, T have 5
(a) both vertical and horizontal lines of symmetry,
(b) rotational symmetry of order 2?

20. A regular polygon has 12 sides. Write its number of lines of symmetry, its order of rotational symmetry and its smallest angle of rotation. Name the polygon. 5

OR

Which of the letters S, I, A, T, H have 5
(a) both vertical and horizontal lines of symmetry,
(b) rotational symmetry of order 2?

END OF QUESTION PAPER Total: 40 Marks

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